

A Convention-Closed Darboux–Halphen Atlas: Theta Series, Modular Covariance, and Collision Strata

Route-A source-local certificate HCS-C320

3 September 2026

Abstract

We lock a polynomial, nome, and theta convention for the Darboux–Halphen system and prove that the logarithmic derivatives of the three Jacobi theta constants solve it. Their exact series also give an Eisenstein divisor-sum bridge. We prove the affine weight-two covariance, its modular-generator permutations, the Chazy reduction, and discriminant transport. We close every pair-collision branch, including coordinate-axis equilibria, and state the full cusp/pole and Route-A boundary.

1 Polynomial convention and theta/q-series lock

Let a prime mean $d/d\tau$ and fix

$$x'_1 = x_2x_3 - x_1(x_2 + x_3), \quad x'_2 = x_3x_1 - x_2(x_3 + x_1), \quad x'_3 = x_1x_2 - x_3(x_1 + x_2). \quad (1)$$

For $\tau \in \mathbb{H}$, set $Q = e^{\pi i \tau}$ and use theta constants with this nome. Define

$$(x_1, x_2, x_3) = -2 \frac{d}{d\tau} (\log \vartheta_2, \log \vartheta_3, \log \vartheta_4). \quad (2)$$

Theorem 1 (Theta seed, exact series, and arithmetic bridge). *The triple (2) solves (1). Put $D = Q d/dQ$ and $X_j = x_j/(\pi i)$. Then $DX_1 = X_2X_3 - X_1(X_2 + X_3)$, cyclically, and*

$$\begin{aligned} \vartheta_2 &= 2Q^{1/4} \sum_{m \geq 0} Q^{m(m+1)}, \\ \vartheta_3 &= 1 + 2 \sum_{m \geq 1} Q^{m^2}, \quad \vartheta_4 = 1 + 2 \sum_{m \geq 1} (-1)^m Q^{m^2}. \end{aligned} \quad (3)$$

In particular $X(0) = (-1/2, 0, 0)$. Moreover

$$X_1 + X_2 + X_3 = -\frac{1}{2}E_2(\tau) = -\frac{1}{2} + 12 \sum_{n \geq 1} \sigma_1(n)Q^{2n}. \quad (4)$$

Proof. Jacobi's triple products give (3). Their logarithmic derivatives converge normally on compact subsets of $|Q| < 1$. The standard theta derivative identities are

$$\begin{aligned} \frac{1}{\pi i} (\log \vartheta_2)' &= \frac{1}{12} (E_2 + \vartheta_3^4 + \vartheta_4^4), \\ \frac{1}{\pi i} (\log \vartheta_3)' &= \frac{1}{12} (E_2 + \vartheta_2^4 - \vartheta_4^4), \\ \frac{1}{\pi i} (\log \vartheta_4)' &= \frac{1}{12} (E_2 - \vartheta_2^4 - \vartheta_3^4). \end{aligned}$$

Substitution, using $\vartheta_3^4 = \vartheta_2^4 + \vartheta_4^4$,

$$E_4 = \frac{1}{2}(\vartheta_2^8 + \vartheta_3^8 + \vartheta_4^8), \quad DE_2 = \frac{1}{6}(E_2^2 - E_4)$$

for the present square-root nome Q , reduces each residual in (1) to zero. Since $d/d\tau = \pi i D$, the scaled polynomial law follows without an omitted factor. Finally $\vartheta_2\vartheta_3\vartheta_4 = 2\eta^3$ and $(\log \eta)' = \pi i E_2/12$ give (4); its last equality is the standard E_2 series in Q^2 . $\square$

Equation (4) is an intrinsic quasimodular divisor-sum identity. It is neither a target Λ -function nor a prime-orbit correspondence.

2 PSL2 covariance, Chazy, and discriminant

Let $F_i(x)$ denote the right side of (1). A direct expansion gives, for a common scalar r ,

$$F_i(w_1 + r, w_2 + r, w_3 + r) = F_i(w) - 2rw_i - r^2. \quad (5)$$

Theorem 2 (Affine weight-two covariance). *If $\gamma \in \text{SL}_2(\mathbb{C})$ has entries a, b, c, d with $ad - bc = 1$ and x solves (1), then, wherever defined,*

$$x_i^\gamma(\tau) = (c\tau + d)^{-2} x_i(\gamma\tau) + \frac{c}{c\tau + d} \quad (6)$$

is another solution. For the theta seed, the modular generators act by

$$T: (x_1, x_2, x_3)(\tau + 1) = (x_1, x_3, x_2)(\tau), \quad (7)$$

$$S: \tau^{-2}(x_1, x_2, x_3)(-1/\tau) + \tau^{-1}(1, 1, 1) = (x_3, x_2, x_1)(\tau). \quad (8)$$

Proof. Write $u = c\tau + d$. Since $(\gamma\tau)' = u^{-2}$, differentiating (6) gives $u^{-4}F_i(x) - 2cu^{-3}x_i - c^2u^{-2}$; equation (5) gives the same expression on its right side. This proves covariance. The standard theta transformations under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$, followed by logarithmic differentiation, give (7)–(8); constant eighth-root multipliers disappear. $\square$

Theorem 3 (Chazy and relative discriminant laws). *For*

$$S = x_1 + x_2 + x_3, \quad \Delta = (x_1 - x_2)(x_2 - x_3)(x_3 - x_1),$$

every solution satisfies

$$S''' = -4SS'' + 6(S')^2, \quad \Delta' = -2S\Delta. \quad (9)$$

Consequently $\Delta = 0$ is an invariant union of collision strata.

Proof. Summing (1) first gives $S' = -(x_1x_2 + x_2x_3 + x_3x_1)$. Differentiate twice and eliminate every x'_i with (1); collecting symmetric monomials yields the first identity. Subtracting two cyclic equations gives, for example, $(x_1 - x_2)' = -2x_3(x_1 - x_2)$. Multiplication of the three corresponding relations proves the second identity and invariance. $\square$

3 Collision strata, cusp/poles, and audit

Theorem 4 (Complete pair-collision branches). *On $x_1 = x_2 = a$, $x_3 = b$, the reduced equations are*

$$a' = -a^2, \quad b' = a^2 - 2ab. \quad (10)$$

For $c, C \in \mathbb{C}$, every local solution with a not identically zero is, on a domain where $\tau \neq c$,

$$a = \frac{1}{\tau - c}, \quad b = \frac{1}{\tau - c} + \frac{C}{(\tau - c)^2}. \quad (11)$$

The layer $C = 0$ is the nonzero fully diagonal Riccati solution. If $a \equiv 0$, then $b \equiv B$ is arbitrary for $B \in \mathbb{C}$, giving the equilibrium axis $(0, 0, B)$. Cyclic permutation gives all collision strata and all three coordinate axes; the axes meet at the origin.

Proof. Substitution gives (10). If a vanishes once, uniqueness for the polynomial ODE forces $a \equiv 0$. Otherwise a Riccati solution is the first formula in (11); its second equation is linear and integration gives the stated C . If $a = 0$, the second equation says $b' = 0$. Cyclic symmetry completes the atlas. $\square$

The theta products are holomorphic and nonvanishing for $\tau \in \mathbb{H}$, so the theta chart is regular there. The face $Q = 0$ is the cusp limit $X = (-1/2, 0, 0)$, not an interior time. In (6), $c\tau + d = 0$ is a chart pole. Theta-zero logarithmic-derivative poles are only a risk after analytic continuation outside $\mathbb{H}$. The invariant collision union, its reciprocal diagonal, and its coordinate-axis equilibria are separate declared layers; no exhaustive classification of all meromorphic continuations is claimed.

Finite evidence. Exact evidence contains 129 coefficient rows through Q^{128} , six high-precision theta/ODE/ S, T rows, 18 reciprocal collision rows, and 15 axis equilibria: 1,705 scalar leaves. A producer-independent Jacobi-product checker performs 1,945 checks, SymPy closes 345 identities, isolated replay is byte exact, and 56 repaired-digest/parser attacks fail. Sampling is a convention lock; the displayed derivations prove the analytic statements.

Registry collision boundary. C186 is the Euler-top Jacobi action–angle flow and C244 the spherical pendulum’s elliptic/focus–focus atlas. C17–C18 concern modular scattering clocks and open traces; C35 is adelic Hénon–theta scattering. None is the present classical three-component modular differential system.

Route-A result and nonclaims. Under NO_BAD_EULER_OR_ROOT_NUMBER, the tuple is

(A0_WEAK_ARITHMETIC_RELATION, A1_FAIL, A2_FAIL,
A3_PARTIAL_ANALYTIC_STRUCTURE, A4_FAIL).

The source owns theta/ E_2 divisor sums and strong analytic covariance, but no target-prime local data, isolated primitive ledger, Euler product, target functional equation, Weil compression, natural self-adjoint quantization, or target-zero match. Route A is rejected and Route B remains locked. No root number, automorphy transfer, or Hilbert–Pólya operator is claimed.

AI use. A generative language model assisted drafting and code scaffolding. The displayed derivations, independent reconstructions, hostile tests, and deterministic artifacts define the audit record.

Source lineage

References

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- [3] J. Harnad and J. McKay, “Modular Invariants and Generalized Halphen Systems,” arXiv: [solv-int/9902012](https://arxiv.org/abs/solv-int/9902012).